Unit ID

SKU:U124











Description

Unit ID is an encryption coprocessor with hardware-based secure key storage, integrating the ATECC608B hardware encryption chip and using the I2C communication interface. The chip has a built-in 10Kb EEPROM, which can be used to store keys, certificates, data, consumption records, and security configurations. By restricting access policies to memory areas, the configuration can be locked to prevent changes.

Each ATECC608B comes with a guaranteed unique 72-bit serial number and includes multiple security features to prevent physical attacks on the device itself or logical attacks on data transmitted between devices.

Supports the Trust&GO platform (pre-configured with universal certificates, suitable for TLS-based network security authentication, such as AWS-IoT, Azure, Google, and other cloud platform verification registrations). The internal certificate of the encryption chip can be directly obtained through tools to complete automatic registration without exposing the private key.

Note

Trust&GO

The pre-configured Trust&GO security element only supports the "Microchip Trust Platform". For more details, please refer to the following link. https://www.microchip.com/en-us/product/ATECC608B-TNGTLS

Features

- ATECC608B-TNGTLSU-G
- I2C communication interface
- Comes with a protective casing for increased stability and reliability
- Protected storage for up to 16 keys, certificates, or data
- Data storage/locking
- Random number generator
- Pre-configured certificates and keys
- 72-bit unique serial number

Includes

- 1 x Unit ID
- 1 x HY2.0-4P Grove Cable (5cm)

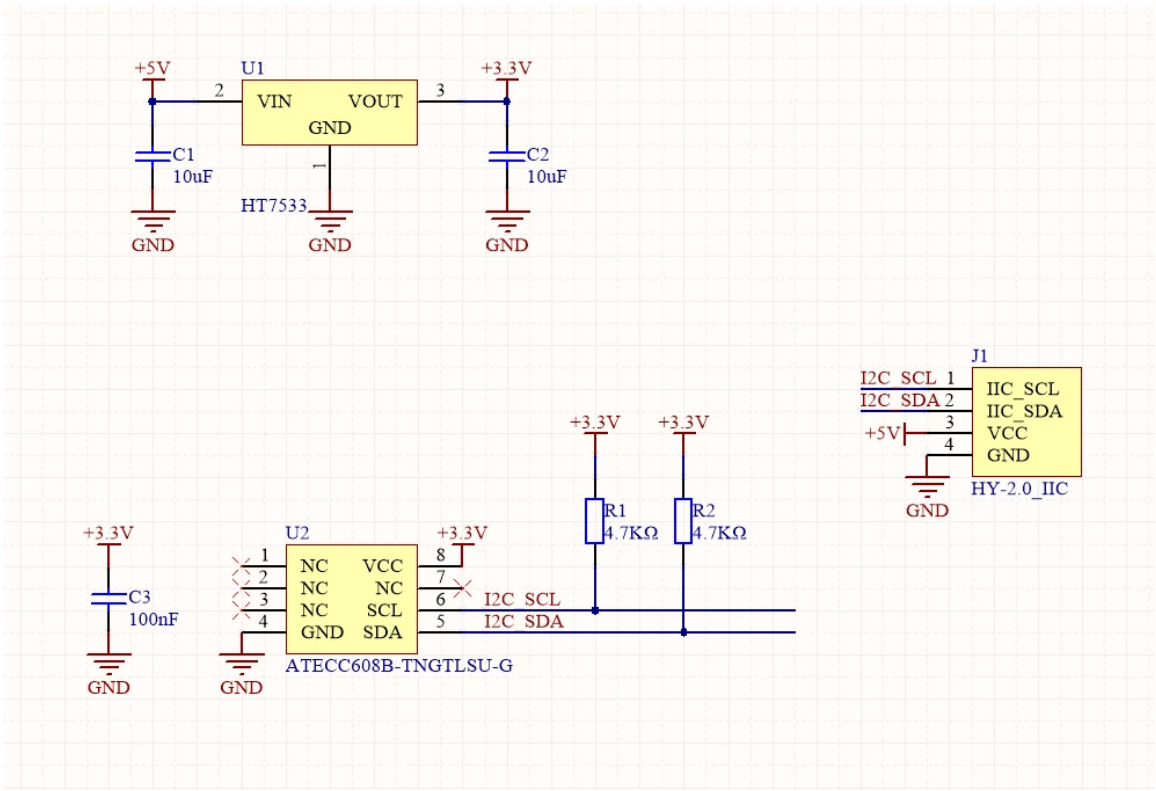
Applications

- IoT endpoint key management and exchange
- Device authentication
- Data encryption

Specifications

Specification	Parameter
Encryption Chip Model	ATECC608B-TNGTLSU-G
Supported Algorithms	ECC-P256 (ECDH and ECDSA), SHA256, AES128-GCM
Supported Key Storage	16
Communication Port	I2C @0x35, maximum bus speed 1Mbps
Power Supply	DC 5V
Product Size	24.0 x 24.0 x 8.0mm
Product Weight	3.3g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	6.5g

Schematics

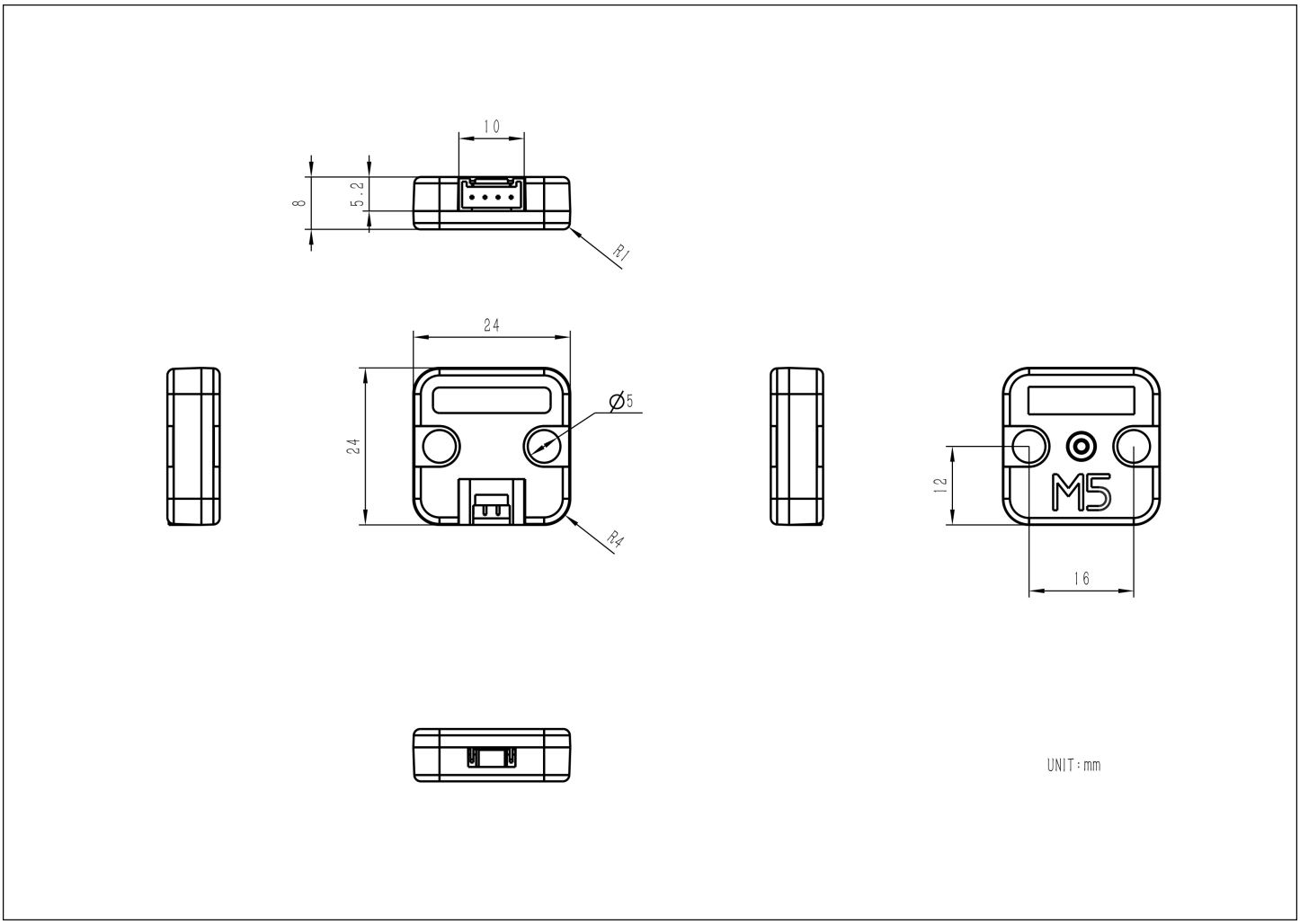


PinMap

Unit ID

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit ID Structure Files

Datasheets

- ATECC608B

Softwares

Arduino

- Unit ID Arduino Library

UiFlow1

- Unit ID UiFlow1 Docs

UiFlow2

- Unit ID UiFlow2 Docs

ESP-IDF

- ECC608-TNG-AWS-Connect

Other

- Generate manifest file